Flamelet-Based Modeling of NO Formation in Turbulent Hydrogen Jet Diffusion Flames

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The potential of using the laminar flamelet model for predictions of NO_x emissions from turbulent hydrogen jet diffusion flames with various amounts of helium dilution is explored in this study. The flamelet approach treats turbulent flames as an ensemble of laminar flames subject to local fluid dynamic stretch which causes chemical kinetics to deviate from equilibrium. This flame stretch rate can be described as the scalar dissipation rate or as the strain rate. Numerical modeling is performed to assess the merits of these two choices by comparing predicted scaling behavior of NO_x emission index with the experimental data of Driscoll and co-workers [1, 2]. The present study reveals that only with the scalar dissipation rate as the nonequilibrium parameter will the predicted NO_x emission indices exhibit a clear scaling relation vs the Damköhler number. The flamelet predictions are found to improve when the Damköhler number increases. However, the absolute NO_x levels are overpredicted, which is attributed to differential diffusion effects.
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INTRODUCTION

Regulations on emissions of nitrogen oxides, NO_x, from combustion processes are becoming more stringent in many countries in efforts to improve urban air quality. Since most practical burners utilize turbulent flows to enhance the combustion rate, it is important to understand the complex interactions between the turbulent flow and the chemical reactions in order to better identify the controlling parameters that lead to reduction in NO_x emissions. The amount of NO_x emission is conventionally quantified by the NO_x emission index $EINO_x$, which is defined as the total mass of NO_x emitted per unit mass of fuel burned. In combustion processes, most NO_x is formed near the reaction zone in the form of nitric oxide, NO. The formation of NO is a relatively slow process compared to the principal combustion reactions. This implies that a large range of chemical time scales, and consequently Damköhler numbers, needs to be properly treated by the turbulent combustion model. The Damköhler number is the ratio between a typical flow time scale and a chemical time scale. Since most turbulent combustion models are developed for a limited regime where the Damköhler number is either large or small, modeling of turbulent combustion with a large disparity of Damköhler numbers is a challenging task. Turbulence–chemistry interactions can be studied most conveniently when chemistry is relatively simple, as in the case of hydrogen combustion. In hydrogen flames at ambient pressure, NO is formed principally by the thermal Zeldovich mechanism. Prompt NO is not generated, and the combustion reactions are well known [3].

Many experiments have been performed in the past to study the characteristics of NO_x emissions from turbulent jet diffusion flames providing the basic understanding of NO_x formation in turbulent flames; see Refs. [1, 2, 4–10] and the review paper by Turns [11]. In particular, the experiments by Chen and Driscoll [1] on free turbulent hydrogen jet flames showed that the $EINO_x$ normalized by a flame residence time scales with $(U_j/D)^{1/2}$, which corresponds to a slope of 0.5 on a log–log plot. The flame residence time is defined as $\tau_u = L_v^3/(D^2 U_j)$,
where L_v is the visible flame length, D is the nozzle diameter, and U_j is the jet exit velocity.

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In the present study, plots of $EINO_x$ vs the Damköhler number $Da = \tau_u/\tau_c$ will be compared. The characteristic chemical time τ_c can be defined as τ_{NO} [12]:

$$\tau_{NO} = \frac{[NO]_{eq}}{S_{NO,eq}} \tag{1}$$

where $[NO]_{eq}$ is the equilibrium NO concentration and $S_{NO,eq}$ is the NO production rate at stoichiometry. In the study of Driscoll et al. [2], the range of Damköhler numbers based on the NO time scale was varied over four decades by adding various amounts of helium to the fuel. For a given dilution, Da increases with D and it decreases with U_j because the normalized flame length L_{st}/D is approximately constant. However, with increasing amount of dilution, L_{st}/D decreases and τ_{NO} increases. These two combined effects cause a large variation in Damköhler numbers.

The scaling behavior of $EINO_x$ has been the subject of many modeling studies in the literature. Early theoretical studies of NO formation in turbulent jet diffusion flames were based on the assumption of chemical equilibrium [6, 13, 14, 15]. Here, a conserved scalar (e.g., the mixture fraction) is introduced to describe the mixing field. With the equal diffusivity assumption, mass fractions of chemical species and the flame temperature are directly related to the conserved scalar. Given the distribution of the conserved scalar, all thermochemical properties can be determined. This fast chemistry approach avoids solving transport equations for mean mass fractions and temperature. These are obtained by using an assumed shape probability density function (PDF) for the mixture fraction. The NO mass fraction is obtained from its own transport equation in which the source term only depends on the mixture fraction. Past studies showed that the NO_x emissions predicted by the fast chemistry model scale linearly with a global flame residence time, which implies a slope of 1. This is not supported by the experiments of Refs. 1, 2, 7, 9, 10, 16, 17.

Attempts at improving the fast chemistry model have been made by invoking the partial equilibrium assumption for the two-body "shuffle" reactions. The overall chemical progress is controlled by the three-body recombination steps, and it is tracked with a reaction progress variable [18]. The chemical source terms are a function of the mixture fraction and the reaction progress variable. This method generates a slope which is much smaller than 0.5 [19]. Apparently, the method is inadequate for describing a large degree of nonequilibrium.

The conditional moment closure (CMC) method does give a slope close to 0.5 [20]. In the CMC method, mean thermochemical quantities conditioned on the mixture fraction are determined for each axial station. Subsequently, the mean values are computed using an assumed shape PDF. The CMC method has been applied to flames with small departure from equilibrium, but it can be used with a detailed chemical mechanism. The CMC method gives absolute levels of $EINO_x$ that are close to the experimental data, except for highly strained flames, where the method underpredicts the $EINO_x$.

A turbulent combustion model that does not rely on the assumption of high Damköhler numbers is the joint PDF transport equation method [21]. A joint PDF of all scalar variables can be solved using a stochastic Monte Carlo method, and the chemical source terms in the transport equations for the mean mass fractions are treated without modeling, in contrast to the mixture-fraction-based methods. Due to the large computational cost of the Monte Carlo method, reduced chemistry has to be used. Also, the term describing molecular diffusion needs modeling, and current models still pose difficulties when the reaction zone is thin. With this constraint, the PDF method is considered more appropriate for chemistry with a broad reaction zone [22, 23]. Chen and Kollmann [24] and Chen et al. [12] applied this method to the flames of Chen and Driscoll, and were able to reproduce the measured 0.5 slope over the whole range of four decades of the Damköhler number when a sufficiently accurate reduced mechanism was used. The absolute $EINO_x$ values are overpredicted by a factor of about 2 for high Damköhler number

flames, which may be due to radiation heat loss.

So far, the potential of using the laminar flamelet method [25, 26] for predicting the $EINO_x$ scaling behavior in turbulent jet flames has not been extensively evaluated. In the flamelet method, the turbulent flame is represented by an ensemble of laminar steady-state stretched one-dimensional flames. The flamelet model is assumed to be applicable when Damköhler numbers are large. Laminar flame calculations with a detailed chemistry are performed to generate a flamelet library in which mass fractions and the temperature are given in terms of the mixture fraction and a parameter that describes the stretch rate. The stretch rate quantifies the state of nonequilibrium of the flamelet. Two parameters currently are in use to quantify the stretch rate: the scalar dissipation rate and the strain rate. The theoretically correct parameter is the scalar dissipation rate [26]. However, recent studies suggest that the strain rate appears to be more appropriate [27–31]. The rationale for using the strain rate, for instance Refs. 27, 30, was that only with this parameter could the lift-off height be predicted with the laminar diffusion flamelet method in turbulent diffusion flames. Mueller et al. [31] included partial premixing in their model, and successfully predicted the lift-off height using the scalar dissipation rate. However, they computed the mean mass fractions still using the strain rate as the nonequilibrium parameter. Consequently, the question of which parameter is responsible for nonequilibrium effects, the scalar dissipation rate or the strain rate, is still open, and it is addressed in the present study.

To compute the mean NO mass fractions in the turbulent flame, the instantaneous NO source terms from the laminar flamelets can be used. In Ref. 32, it was shown semianalytically that using the strain rate as the nonequilibrium parameter, the 0.5 slope would not be reproduced with the flamelet method. Moreover, using the strain rate, the slope of the curve would increase with respect to the equilibrium value of 1. This was concluded from flow and temperature field similarity in high Reynolds and high Froude number flames in combination with the behavior of the maximum NO production rate as a function of the strain rate in laminar flamelets. In the present study, complete flame calculations are presented in which the scalar dissipation rate, the small eddy strain rate, and the large eddy strain rate are used as the mean nonequilibrium parameter. Computations are performed for the flames of Driscoll and co-workers [1, 2], who investigated pure and helium diluted hydrogen flames with various amounts of helium from 20 to 70% by volume. The fuel jet nozzle diameters used were $d = 1.6$, 2.6, and 3.75 mm, and the calculations are done without gravity and radiation effects, which are believed of secondary importance for turbulent hydrogen jet flames at a high Froude number. With these calculations, the effect of choosing different nonequilibrium parameters on the predicted slope of the $EINO_x$ curve will be explored. Two types of implementations of the flamelet method are compared: one in which transport equations for the mean mixture fraction and its variance are solved, and one in which a PDF transport equation for the mixture fraction is solved using the Monte Carlo method. The latter model was used by Chen et al. [12] with the joint PDF approach that reproduces the 0.5 slope, as mentioned earlier. Implementation of the flamelet model into the same stochastic model is useful for comparison with the traditional flamelet model. Such a comparison will indicate whether or not the predicted slope is sensitive to the implementation of the turbulent combustion model.

In the next section, the $EINO_x$ scaling behavior in turbulent hydrogen flames is analyzed, and it is related to the characteristics of NO formation in laminar flamelets. From this analysis, it is tentatively concluded that the strain rate would not be the nonequilibrium parameter that is able to reproduce the experimental 0.5 slope of $EINO_x$ vs Da. Subsequent detailed flame calculations, using the two implementations of the flamelet method, are presented to confirm the previous analysis. The impact of differential diffusion effects on the predicted slope and on the absolute value of $EINO_x$ is discussed, and finally, the conclusions are presented.

ANALYSIS OF THE SCALING BEHAVIOR OF THE EMISSION INDEX

The emission index EINO_x is conventionally defined as

$$\mathrm{EINO}_x = \mathrm{EINO}\frac{M_{\mathrm{NO}_2}}{M_{\mathrm{NO}}} + \mathrm{EINO}_2 \tag{2}$$

where M_i are the molecular masses. As NO_2 is produced from NO, it is only necessary to model NO formation in order to determine EINO_x from hydrogen flames. The definition of EINO is the total mass of NO produced in the flame per kg of fuel:

$$\mathrm{EINO} = \frac{2\pi D^3 \int_0^\infty d\left(\frac{x}{D}\right)\int_0^\infty \frac{r}{D}\bar{\dot{w}}_{\mathrm{NO}}\, d\left(\frac{r}{D}\right)}{\frac{1}{4}\pi\rho_f U_j D^2 Y_f}$$

$$= \frac{8}{\rho_f}\frac{D}{U_j Y_f} I \tag{3}$$

where ρ_f is the density of the nozzle gas (fuel) and Y_f is the fuel mass fraction in the fuel stream. Here, the integral I is proportional to $I \sim (1/D^3)\iiint \bar{\dot{w}}_{\mathrm{NO}}\, dV$. The effect of nonequilibrium chemistry is contained in the term $\bar{\dot{w}}_{\mathrm{NO}}$. The main contribution to the integral I comes from the region where the mean centerline mixture fraction is near stoichiometric [20, 32]. Therefore, the mean value of the nonequilibrium parameter in the stoichiometric region determines, to a large extent, the amount of NO formed in the flame. If combustion chemistry is assumed to be at equilibrium, I is constant when the flames obey Reynolds similarity [32]. Similarity only exists without coflow, and if either the Froude number is constant or there is no gravity force. When similarity is established, the profiles of all variables normalized by appropriate powers of U_j and D are identical and independent of U_j and D. Hence, assuming similarity and chemical equilibrium, I will be constant. The flames investigated by Chen and Driscoll are believed to satisfy the similarity criteria, but their experiments showed that $\mathrm{EINO}_x \sim (D/U_j)^{1/2}$, and hence, $I \sim (U_j/D)^{1/2}$. This implies that the equilibrium chemistry assumption, in which case I would be constant and $\mathrm{EINO}_{x,\mathrm{eq}} \sim D/U_j$, is not valid. Note that the ratio U_j/D can be interpreted as a global flame strain rate.

As argued above, the main contribution to I comes from the region where the mean mixture fraction is around stoichiometric. This is consistent with the fact that the NO production rate in each laminar flamelet is maximal around the stoichiometric mixture fraction. Therefore, the integral I is approximately proportional to $I \sim \dot{w}_{\mathrm{NO,max}}(\zeta = \tilde{\zeta}_{c,st})$, where $\tilde{\zeta}_{c,st}$ is the mean centerline value of the nonequilibrium parameter at the axial location where the mean mixture fraction is stoichiometric. The question is whether or not this leads to the (experimental) scaling relation $I \sim (U_j/D)^{1/2}$. If $\tilde{\zeta}$ is associated with either the mean scalar dissipation rate ϵ_f or with the large eddy strain rate ϵ/k, then $\tilde{\zeta} \sim U_j/D$ [32]. Therefore, $\dot{w}_{\mathrm{NO,max}}(\zeta = \tilde{\zeta}_{c,st})$ needs to be an increasing function of $\tilde{\zeta}_{c,st}$, according to the experiments. This implies that $\dot{w}_{\mathrm{NO,max}}(\zeta)$ should be an increasing function of ζ for values of $\zeta \approx \tilde{\zeta}_{c,st}$ in the turbulent flames considered here. This is a remarkable constraint since, in general, the NO production rate decreases with increasing strain. This is due to the decrease of temperature with the strain rate in adiabatic flames. However, the NO production rate also depends on the oxygen radical concentration which can modify this behavior. In the pure hydrogen–air flamelet library, $\dot{w}_{\mathrm{NO,max}}$ exhibits a maximum at $\zeta_{\max}$, as presented in Fig. 1. This maximum is due to the competitive effect between temperature and the oxygen radical concentration in the most important, rate-limiting, NO-producing reaction $N_2 + O \rightarrow NO + N$:

$$\dot{w}_{\mathrm{NO}} \sim C_{\mathrm{O}} C_{\mathrm{N}_2} \exp(-T_a/T) \tag{4}$$

where C_{O} and C_{N_2} are the oxygen radical and nitrogen molecule concentrations, respectively, and T_a is the activation temperature of about 38,000 K. The maximum temperature in the flamelets decreases with increasing strain rate. This temperature decrease reduces the NO production substantially at high strain rates due to the high activation temperature. However, as can be seen in Fig. 1, the maximum oxygen radical concentration increases with the

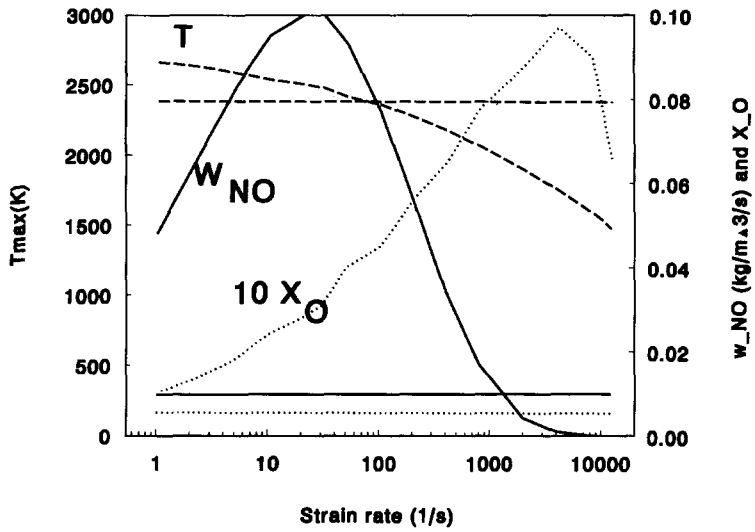

Fig. 1. Maximum NO production rate w_{NO} is kg/m^3, maximum temperature, and O radical mole fraction vs strain rate in pure H_2–air flamelets.

strain rate; only near extinction does the O atom concentration start to decrease. High oxygen radical concentrations favor NO production. The dependence of NO production rate on the strain rate is a result of these two competing effects, which lead to a maximum of the NO production rate at ζ_{max} as shown in Fig. 1. For pure hydrogen flames, ζ_{max} corresponds to a value of the strain rate $s_{max} \approx 25$ s^{-1} and a value of the scalar dissipation rate at stoichiometric mixture $\chi_{max} \approx 0.1$ s^{-1}. Note that instantaneous values of the scalar dissipation rate are denoted by χ, while the mean value is denoted by ϵ_f. Now, the nonequilibrium parameter ζ needs to be identified. It can be defined either as the scalar dissipation rate or as the strain rate. Subsequently, the mean value of this parameter in the turbulent flame must be specified. The mean strain rate can be determined either by the large eddy strain rate $\tilde{s}_l$ or by the small eddy strain rate $\tilde{s}_s$. If $\tilde{\zeta}_{c,st} < \zeta_{max}$, the integral I will increase with $\tilde{\zeta}$, and therefore it will increase with U_j/D, which is a necessary condition to reproduce the experimental trend. Here, fluctuations in ζ are neglected for simplicity. In Ref. 32, it was shown that when ζ is assigned as the strain rate, the integral I decreases with U_j/D, which is opposite to the observed trend. This is because all mean strain rates $\tilde{s}_{c,st}$ were larger than the strain rate s_{max} in the main NO production regions of the flames investigated by Chen and Driscoll. The strain rate and the scalar dissipation rate differ in their scaling behaviors in terms of downstream distance. In a turbulent jet, the strain rate scales as x^{-2} while the scalar dissipation rate scales as x^{-4}. Unlike the strain rate, the mean scalar dissipation rate may decrease well below χ_{max} in the main NO production region. Under this condition, I increases with ϵ_f, and therefore it also increases with U_j/D, as observed experimentally. Whether or not the 0.5 slope can be reproduced by the flamelet approach will be investigated by a series of numerical simulations over seven decades of the Damköhler number.

NUMERICAL MODELS

The turbulent flames are computed using two numerical codes, called code A and code B, which differ in the second-order turbulence model and the implementation of the laminar flamelet model. Transport equations for mean velocity, mixture fraction, Reynolds stresses, scalar fluxes, and the kinetic energy dissipation rate are solved for turbulent jet diffusion flames using the parabolic flow assumption.

All variables, except the density and pressure, are Favre-averaged (mass-weighted) to prevent the appearance of too many density

fluctuation correlation terms. Favre-averaged variables are either denoted by a tilde $\tilde{\phi}$ or simply by capitals. Conventional averages are denoted by the overbar $\bar{\phi}$. Favre fluctuations are denoted by ϕ'', while conventional fluctuations are indicated by ϕ'. The continuity equation is given by

$$\frac{\partial \bar{\rho} U}{\partial x} + \frac{1}{r}\frac{\partial r \bar{\rho} V}{\partial r} = 0 \tag{5}$$

where x is the axial distance in the direction of the jet and r is the radial distance. The equation for axial momentum is

$$\frac{\partial \bar{\rho} UU}{\partial x} + \frac{1}{r}\frac{\partial r \bar{\rho} UV}{\partial r} = -\frac{\partial \bar{p}}{\partial x} - \frac{1}{r}\frac{\partial r \bar{\rho}\widetilde{u''v''}}{\partial r} + \text{Mol.Dif.} \tag{6}$$

where the molecular diffusion term is designated by Mol.Dif. Buoyancy effects are omitted, and the mean pressure gradient is assumed 0.

The mixture fraction F obeys a convection–diffusion equation of the form

$$\frac{\partial \bar{\rho} UF}{\partial x} + \frac{1}{r}\frac{\partial r \bar{\rho} VF}{\partial r} = \frac{1}{r}\frac{\partial r \bar{\rho}\widetilde{v''f''}}{\partial r} + \text{Mol.Dif.} \tag{7}$$

The Reynolds stress equations are

$$\frac{\partial}{\partial x_k}\left(\bar{\rho} U_k \widetilde{u_i'' u_j''}\right) = P_{ij} + D_{ij} + \overline{p'\left(\frac{\partial u_j''}{\partial x_i} + \frac{\partial u_i''}{\partial x_j}\right)} - \frac{2}{3}\delta_{ij}\bar{\rho}\epsilon \tag{8}$$

in which local isotropy in the smallest scales has been assumed to model the dissipation term (last term). The first term on the right-hand-side is the production term due to mean strain, which needs no modeling:

$$P_{ij} = -\bar{\rho}\left(\widetilde{u_i'' u_k''}\frac{\partial U_j}{\partial x_k} + \widetilde{u_j'' u_k''}\frac{\partial U_i}{\partial x_k}\right). \tag{9}$$

The diffusion (D_{ij}) and pressure strain terms in Eq. 8 need modeling. Equation 8 is solved for $\widetilde{u''u''}$, $\widetilde{v''v''}$, and $\widetilde{u''v''}$. The tangential fluctuations $\widetilde{w''w''}$ are assumed equal to $\widetilde{v''v''}$. Now, the turbulence and combustion models in both codes will be described.

Model *A*

The diffusion term in Eq. 8 is modeled as

$$D_{ij} = C_s \frac{\partial}{\partial x_k}\left[\left(\bar{\rho}\frac{k}{\epsilon}\widetilde{u_k'' u_l''} + \bar{\rho}\,\delta_{kl}\nu\right)\frac{\partial \widetilde{u_i'' u_j''}}{\partial x_l}\right] \tag{10}$$

where k is the turbulent kinetic energy per unit mass:

$$k = \tfrac{1}{2}\widetilde{u_i'' u_i''} \tag{11}$$

using the summation convention and ν is the molecular kinematic viscosity.

The pressure–strain correlation is

$$\overline{p'\left(\frac{\partial u_j''}{\partial x_i} + \frac{\partial u_i''}{\partial x_j}\right)} = \phi_{ij,1} + \phi_{ij,2}. \tag{12}$$

In code *A*, the Launder–Reece–Rodi (LRR) model [33] is used for this term. The "return to isotropy" term $\phi_{ij,1}$ is expressed in the anisotropy tensor b_{ij} which is defined as

$$b_{ij} = \frac{\widetilde{u_i'' u_j''}}{2k} - \frac{1}{3}\delta_{ij}. \tag{13}$$

The return term thus becomes

$$\phi_{ij,1} = -C_1\,\bar{\rho}\epsilon 2 b_{ij}. \tag{14}$$

The mean strain, also called the "rapid term," is modeled as

$$\phi_{ij,2} = -\alpha\left(P_{ij} - \tfrac{2}{3}\delta_{ij}P\right) \tag{15}$$

where the production P of turbulent kinetic energy is

$$P = \tfrac{1}{2}P_{kk}. \tag{16}$$

The dissipation rate equation is exactly the same as in the standard k-ϵ model, except for the diffusion term, which in the Reynolds stress model is the same as in Eq. 10 with a coefficient $C_{\epsilon,s}$ instead of C_s.

The scalar turbulence equations are given by

$$\frac{\partial}{\partial x_k}\left(\bar{\rho}U_k\widetilde{u_i''f''}\right) = P_i + D_i + \overline{p'\frac{\partial f''}{\partial x_i}} \tag{17}$$

with shear production defined as

$$P_i = -\bar{\rho}\left(\widetilde{u_k''f''}\frac{\partial U_i}{\partial x_k} + \widetilde{u_i''u_k''}\frac{\partial F}{\partial x_k}\right). \tag{18}$$

The diffusion term is modeled as

$$D_i = C_{sf}\frac{\partial}{\partial x_k}\left(\bar{\rho}\frac{k}{\epsilon}\widetilde{u_k''u_l''}\frac{\partial\widetilde{u_i''f''}}{\partial x_l}\right). \tag{19}$$

The pressure–scalar gradient correlation is defined by

$$\overline{p'\frac{\partial f''}{\partial x_i}} = \phi_{i,1} + \phi_{i,2} \tag{20}$$

in which the return term is modeled as

$$\phi_{i,1} = -C_{1f}\bar{\rho}\frac{\epsilon}{k}\widetilde{u_i''f''} - C_{1f}'\bar{\rho}\frac{\epsilon}{k}b_{ik}\widetilde{u_k''f''}. \tag{21}$$

The mean strain or rapid term is modeled as

$$\phi_{i,2} = 0.8\bar{\rho}\widetilde{u_k''f''}\frac{\partial U_i}{\partial x_k} - 0.2\bar{\rho}\widetilde{u_k''f''}\frac{\partial U_k}{\partial x_i}. \tag{22}$$

From the foregoing, we can deduce the parabolized set of equations in cylindrical coordinates where the generalized equation is

$$\frac{\partial}{\partial x}\left(\bar{\rho}U\tilde{\phi}\right) + \frac{1}{r}\frac{\partial}{\partial r}\left(r\bar{\rho}V\tilde{\phi}\right) = \frac{1}{r}\frac{\partial}{\partial r}\left(r\bar{\rho}\mathscr{D}\frac{\partial\tilde{\phi}}{\partial r}\right) + S_\phi \tag{23}$$

with S_ϕ given in Table 1, $P = -\bar{\rho}\widetilde{u''v''}(\partial U/\partial r)$. The model constants used in code A are given in Table 2.

The scalar dissipation rate is obtained from the algebraic relationship

$$\epsilon_f = 2\widetilde{f''^2}\frac{\epsilon}{k} \tag{24}$$

TABLE 1

Source Terms for the Second-Order Model in Code A in the Generalized Equation 23. The "Turbulent Diffusion" Coefficient D is $C_{\text{diff}}(k/\epsilon)\widetilde{v''v''}$. For All Reynolds stresses $C_{\text{diff}} = C_s$, for the Scalar Fluxes and the Scalar Variance $C_{\text{diff}} = C_{sf}$, and for the Dissipation Rate $C_{\text{diff}} = C_{\epsilon,s}$.

Variable	S_ϕ
$\widetilde{u''u''}$	$-2(1-\alpha)\bar{\rho}\widetilde{u''v''}\frac{\partial U}{\partial r} - \frac{2}{3}\bar{\rho}\epsilon - C_1\bar{\rho}\frac{\epsilon}{k}(\widetilde{u''u''} - \frac{2}{3}k) + \frac{2}{3}\alpha P$
$\widetilde{v''v''}$	$-\frac{2}{3}\bar{\rho}\epsilon - C_1\bar{\rho}\frac{\epsilon}{k}(\widetilde{v''v''} - \frac{2}{3}k) + \frac{2}{3}\alpha P - \frac{2C_s}{r^2}\frac{k}{\epsilon}\bar{\rho}\widetilde{w''w''}(\widetilde{v''v''} - \widetilde{w''w''})$
$\widetilde{w''w''}$	$-\frac{2}{3}\bar{\rho}\epsilon - C_1\bar{\rho}\frac{\epsilon}{k}(\widetilde{w''w''} - \frac{2}{3}k) + \frac{2}{3}\alpha P + \frac{2C_s}{r^2}\frac{k}{\epsilon}\bar{\rho}\widetilde{w''w''}(\widetilde{v''v''} - \widetilde{w''w''})$
$\widetilde{u''v''}$	$-(1-\alpha)\bar{\rho}\widetilde{v''v''}\frac{\partial U}{\partial r} - C_1\rho\frac{\epsilon}{k}\widetilde{u''v''} - \frac{C_s}{r^2}\frac{k}{\epsilon}\bar{\rho}\widetilde{w''w''}\widetilde{u''v''}$
ϵ	$\frac{\epsilon}{k}(C_{\epsilon,1}P - C_{\epsilon,2}\bar{\rho}\epsilon)$
$\widetilde{u''f''}$	$-\bar{\rho}\widetilde{v''f''}\frac{\partial U}{\partial r} - \bar{\rho}\widetilde{u''v''}\frac{\partial F}{\partial r} - C_{1f}\bar{\rho}\frac{\epsilon}{k}\widetilde{u''f''} - C_{1f}'\bar{\rho}\frac{\epsilon}{k}(b_{uu}\widetilde{u''f''} + b_{uv}\widetilde{v''f''}) + 0.8\bar{\rho}\widetilde{v''f''}\frac{\partial U}{\partial r}$
$\widetilde{v''f''}$	$-\bar{\rho}\widetilde{v''f''}\frac{\partial V}{\partial r} - \bar{\rho}\widetilde{v''v''}\frac{\partial F}{\partial r} - C_{sf}\bar{\rho}\frac{k}{\epsilon}\frac{\widetilde{w''w''}}{r^2}\widetilde{v''f''} - C_{1f}\bar{\rho}\frac{\epsilon}{k}\widetilde{v''f''}$ $-C_{1f}'\bar{\rho}\frac{\epsilon}{k}(b_{vu}\widetilde{u''f''} + b_{vv}\widetilde{v''f''}) - 0.2\bar{\rho}\widetilde{u''f''}\frac{\partial U}{\partial r}$
$\widetilde{f''^2}$	$-2\bar{\rho}\widetilde{v''f''}\frac{\partial F}{\partial r} - \bar{\rho}\epsilon_f$

TABLE 2

Model Coefficients in the Reynolds Stress Model of Code A. The Constants for the Velocity Turbulence, except C_1, are from Ref. 37, and the Constants in the Scalar Turbulence are from Ref. 38. Coefficient C_1 has been Adapted to have a Correct Round Jet Spreading Rate, and the Above Values Correspond to "Model RSM II" in Ref. 34.

C_1	α	C_s	$C_{\epsilon,1}$	$C_{\epsilon,2}$	$C_{\epsilon,s}$	C_{1f}	C'_{1f}	C_{sf}
2.3	0.6	0.22	1.45	1.90	0.18	5.7	−6.1	0.22

using a constant time scale ratio of 2 between scalar and mechanical fluctuations [35]. Transport equations for ϵ_f have also been implemented in code A, but they do not change the predicted slope of the EINO_x curve.

The Favre-averaged mean of a thermochemical variable ϕ, such as ρ, T, Y_i, or $\bar{\dot{w}}_{\mathrm{NO}}$ (but not $\tilde{Y}_{\mathrm{NO}}$), is obtained by

$$\tilde{\phi} = \int_0^1 df \int_0^\infty d\zeta\, \tilde{P}(f,\zeta)\phi(f,\zeta);$$
$$(\bar{\rho})^{-1} = \int_0^1 df \int_0^\infty d\zeta\, \tilde{P}(f,\zeta)/\rho(f,\zeta). \tag{25}$$

Here, $\tilde{P}(f,\zeta)$ is the joint PDF of the mixture fraction f and ζ is the stretch rate. The joint PDF $P(f,\zeta)$ is approximated as $P(f)P(\zeta)$ assuming statistical independence between f and ζ. $P(f)$ is modeled by a β function with its two parameters determined by the first two moments of the mixture fraction, namely $\tilde{f}$ and $\widetilde{f''^2}$, both of which are obtained from their respective transport equations. $P(\zeta)$ is assumed to have a lognormal distribution when ζ equals the scalar dissipation rate χ. Local flame extinction is not considered in code A. Flame extinction is considered unimportant for NO_x emissions because the largest contribution to the NO levels comes from the downstream regions where local extinction is not expected.

Model *B*

Code B differs from code A in the turbulence model and in the solution algorithm of the mixture fraction field. The turbulence model is also a second-order model, based on the work of Lumley [36], and it is described in Ref. 39. In this model, nondimensional groups consisting of the anisotropy tensor and its principal (nonzero) invariants (II and III) and the turbulence Reynolds number $\mathrm{Re}_t = 4k^2/(9\epsilon\nu)$ are introduced. Then, the pressure–strain correlation is modeled with these variables as parameters. The principal differences with model A are the following. The coefficient C_1 is given by

$$C_1 = 1 + 0.5\exp\left(-7.77/\mathrm{Re}_t^{1/2}\right) \tag{26}$$

$$\left[72/\mathrm{Re}_t^{1/2} + 80.1\ln(1 + 62.4(-\mathrm{II} + 2.3\mathrm{III}))\right] \times (1/9 + 9\mathrm{II} + 27\mathrm{III}) \tag{27}$$

with II and III defined as

$$\mathrm{II} = \tfrac{1}{2}\left(b_{ii}b_{jj} - Tr\left(b_{ij}^2\right)\right) = -\tfrac{1}{2}Tr\left(b_{ij}^2\right) = -\tfrac{1}{2}b_{ij}b_{ji} \tag{28}$$

$$\mathrm{III} = \frac{1}{3!}\left(b_{ii}b_{jj}b_{kk} - 3b_{ii}Tr\left(b_{ij}^2\right) + 2Tr\left(b_{ij}^3\right)\right) = \frac{2}{3!}Tr\left(b_{ij}^3\right) = \frac{1}{3}b_{ij}b_{jk}b_{ki} \tag{29}$$

where $b_{ij}^2 = b_{ik}b_{kj}$ and $b_{ij}^3 = b_{ik}b_{kl}b_{lj}$.

The rapid term is

$$\phi_{ij,2} = 2\frac{\partial U_p}{\partial x_q}\left(X_{pj}^{qi} + X_{pi}^{qj}\right). \tag{30}$$

Here, X_{pj}^{qi} is split into an isotropic and an anisotropic part with

$$X_{pj,\mathrm{iso}}^{qi} = \frac{2k}{30}(4\delta_{pj}\delta_{iq} - \delta_{ip}\delta_{jq} - \delta_{pq}\delta_{jp}) \tag{31}$$

$$X_{pj,\mathrm{anis}}^{qi} = C_{ra}\left(\delta_{iq}b_{pj} + \frac{4}{11}\delta_{pj}b_{iq} - \frac{3}{11}Y_{pj}^{qi}\right) + \frac{2k}{11}\left(5\delta_{pj}b_{iq} - Y_{pj}^{qi}\right) \tag{32}$$

with

$$Y_{pj}^{qi} = \delta_{ip} b_{jq} + \delta_{pq} b_{ij} + \delta_{ij} b_{pq} + \delta_{jq} b_{ip} \quad (33)$$

and $C_{ra} = 0.55k$.

The source term S_ϕ in the generalized equation 23 for the dissipation rate ϵ is

$$S_\phi = -\frac{\epsilon^2}{2k}\left\{\Psi_0 + \Psi_1 b_{ij}\frac{2k}{\epsilon}\frac{\partial U_i}{\partial x_j}\right\} \quad (34)$$

with

$$\Psi_0 = 2.8 + 0.98\exp\left(-2.83\,\mathrm{Re}_t^{-1/2}\right) \times (1 - 0.33\ln(1 - 55\mathrm{II})) \quad (35)$$

and $\Psi_1 = 2.42$.

For the mixture fraction field, a PDF transport equation is solved using a Monte Carlo method. This means that no equations for the mean mixture fraction, mixture fraction variance, or scalar fluxes are solved. In each grid cell, an ensemble of statistics of the mixture fraction is used to describe the mixing field. The mean scalar dissipation rate is modeled by a transport equation, and its PDF is a lognormal distribution. The mean thermochemical variables are determined by

$$\overline{\phi} = \frac{1}{N_p}\sum_{i=1}^{N_p} \phi(f_i, \zeta_i) \quad (36)$$

where N_p is the number of statistical "particles" and $\phi(f_i, \zeta_i)$ is obtained using interpolation of the data $\phi(f, \zeta)$ in the flamelet library.

Chemistry

Despite the many differences in codes *A* and *B*, both models are based on second-order moment closure approaches. When combined with a joint scalar PDF approach, code *B* reproduces the observed 0.5 slope [24]. Substitution of the joint scalar PDF model in code *B* by the flamelet approach allows us to assess the sensitivity of the predicted NO_x trend on a different implementation of the turbulent combustion model. To this end, the same flamelet library is used in both codes. The flamelet library is generated by calculating opposed planar jet flames at various strain rates using a computer code that is a precursor of Rogg's RUN1DL [40]. The detailed chemical mechanism used here is given in Ref. 3. The strain rate is evaluated at the oxidizer side, and the scalar dissipation rate is determined at the stoichiometric point. The latter is defined as $\chi = 2D(\partial f/\partial x_j)(\partial f/\partial x_j)$ with D being the mixture diffusivity. Chemical equilibrium is used when ζ approaches 0. The equilibrium values are obtained using the Chemkin PSR code [41]. Boundary conditions correspond to pure fuel and pure oxidizer, both at $T = 300$ K. This corresponds to the classical flamelet approach [26]. Bish and Dahm [44] consider flamelets with boundary conditions on the rich side that are different from the pure fuel case, in view of the fact that in the far field of a jet flame, no pure fuel at 300 K exists. An alternative would be to take equilibrium values for mass fractions and temperature at the rich side (a mix of fuel and hot products). However, this would be equivalent to the present approach using unity Lewis numbers in the flamelets.

When the scalar dissipation rate in the turbulent flame calculations is used as the stretch parameter, the value χ_{st}, i.e., the stoichiometric scalar dissipation rate in the laminar flamelets, corresponds to the instantaneous scalar dissipation rate in the turbulent flame. If, of the other hand, the strain rate is used, the strain rate at the oxidizer side in the laminar flamelets is associated with the instantaneous strain rate in the turbulent flame.

RESULTS AND DISCUSSIONS

A series of exploration runs have been conducted to evaluate the differences in the predicted results from the two codes, and a brief discussion is provided here. Unlike code *B*, local flame extinction is not considered in code *A*. When local extinction is not incorporated into code *B*, the predicted emission indices change insignificantly for the 60% helium-diluted flames in which flame extinction is most likely to occur. This suggests that possible flame extinction has little effect on the predicted NO_x emissions under the flame conditions studied by Chen and Driscoll.

The effect of modeling the distribution of the scalar dissipation rate is considered next. Two PDF shapes are explored. The first one simply assumes a delta function at the mean scalar dissipation rate, i.e., there is no fluctuation. The second one assumes a lognormal PDF with variance determined by $\sigma = 1$. Compared to the results obtained with the delta function, the predicted NO_x emissions decrease by about 14% with code A, and by 4% with code B when the lognormal distribution is used. The predictions with code A with the two PDFs are plotted in Fig. 2. Judging from the differences being less than 15%, we will not include the lognormal distribution in all calculations with code A when performing parametric studies to save time. However, a series of runs, but with limited changes in the parameters, has been conducted using code B with the lognormal distribution, and the results will be compared to those from code A.

Emission Index

Predictions with the Scalar Dissipation Rate

Figure 2 shows the predicted and measured emission indices for all diameters and helium dilutions against the Damköhler number $\mathrm{Da} = (L_{st}/D^*)^3(D^*/U_j\tau_{\mathrm{NO}})$ where L_{st} is the "flame length" defined as the axial distance at which the mean mixture fraction is stoichiometric and D^* is an effective diameter

$$D^* = D\sqrt{\rho_j/\rho_a}$$

where ρ_j and ρ_a are the fuel jet and air density, respectively. The solid line in Fig. 2 denotes a fitted correlation,

$$\mathrm{EINO}_x = 0.139\,\mathrm{Da}^{1/2},$$

based on the measurements by Driscoll and his co-workers [1, 2]. Since only visible flame heights were measured, we have converted the visible flame heights L_v into stoichiometric flame heights L_{st} with the empirical relationship $L_{st}/L_v = 0.75$ from the experimental observation by Barlow and Carter [10].

From the above scaling analysis of EINO_x vs Damköhler number, it was concluded that when equilibrium chemistry is assumed, EINO_x scales linearly with Damköhler number. As seen in the logarithmic plot in Fig. 2, the predictions with equilibrium chemistry using code A for different conditions indeed collapse onto one curve with a slope of unity. Next, we consider the slope of EINO_x vs Damköhler number predicted with the flamelet model. If the mean value of the nonequilibrium parameter near the mean stoichiometric is larger (smaller) than $\zeta_{\max}$, the slope will be

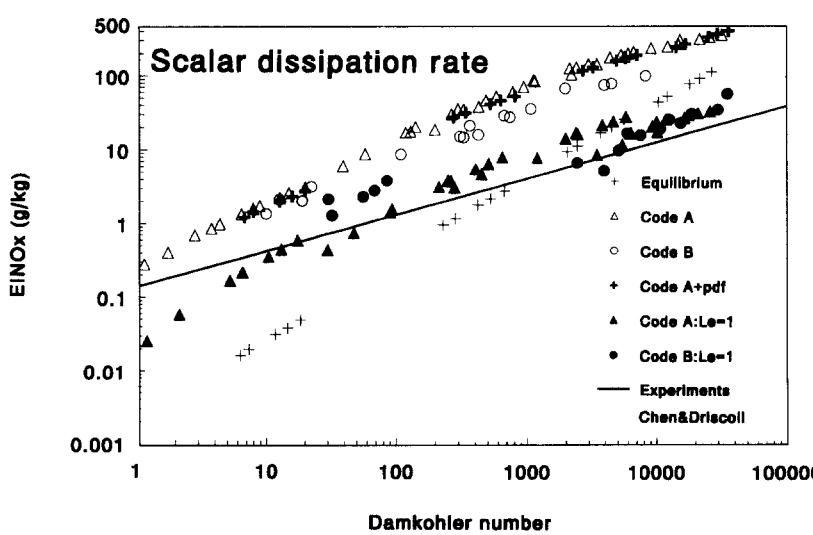

Fig. 2. Emission index vs Damköhler number, predicted with the scalar dissipation rate. The Damköhler number is defined as $\mathrm{Da} = (L/D^*)^3(D^*/U\tau_{\mathrm{NO}})$. The solid line is a fit through the measurements of Chen and Driscoll [1].

greater (smaller) than 1. With the scalar dissipation rate as the nonequilibrium parameter, the flamelet predictions using both codes A and B show that the predicted emission indices do collapse approximately on one curve as seen in Fig. 2. The predicted slope is smaller than 1 over seven decades of Da numbers. This means that the effect of nonequilibrium chemistry, when parameterized with the scalar dissipation rate, shows the same trend as seen in experiments. Closer examination shows that the predicted slope approaches the experimentally observe value of 0.5 for large Da cases, but it increases at low Da conditions.

As shown in Fig. 2, the differences between the predictions with codes A and B are relatively small. This is remarkable given the many differences between the two codes in turbulence modeling, as well as in the implementation of the flamelet model. The differences in the absolute values of $EINO_x$ between the flamelet model and the experiments are quite large. Three possible causes are considered here. On possibility is associated with the uncertainty in predicting the scalar dissipation rate. A test calculation with code A is performed to study the sensitivity of NO_x prediction on the scalar dissipation rate. In this calculation, the flamelet stretch parameter ζ was reduced by a factor of 10, but with the mean scalar dissipation rate kept unchanged in the equation for scalar variance. This decreases $EINO_x$ by a factor of 3, which is still too high compared to the experimental value. This tenfold decrease in the scalar dissipation rate would lead to significant changes in the predicted scalar variance, which is well predicted. Therefore, changing the scalar dissipation rate alone does not explain the large discrepancy between model predictions and measurements. The second possibility is thermal radiation, which leads to lower flame temperatures and consequently lower NO production rates. A radiation model has been used in code B which uses the optically thin limit approximation, and which therefore gives the upper limit regarding radiation effects. Typical reductions in $EINO_x$ for pure and He-diluted hydrogen flames are: 48% for a pure flame, 31% for a 20% He flame, and 8% for a 40% He flame. All flames correspond to Barlow and Carter's experiment [10] at $D = 3.7$ mm and $U_j = 297$ m/s for the pure and 20% He flames and $U_j = 257$ m/s for the 40% He flame. According to these upper limits, radiation can only account for a difference of a factor of 2 in $EINO_x$, while a factor of 10 has to be accounted for. The third possible cause is differential diffusion among species and heat, which is known to cause superequilibrium temperatures in laminar hydrogen flames. This is due to the high diffusivity of H_2 molecules which leads to a higher diffusion flux of fuel (H_2) to the flame zone compared to a hypothetical case where all diffusivities are equal. However, experiments of turbulent hydrogen jet flames indicate that these differential diffusion effects diminish with axial distance as well as with jet exit velocity [10, 42], and therefore also with the strain rate. The effect of differential diffusion in the flamelet library is nearly invariant with the strain rate, as reported by Vranos et al. [43]. Therefore, at downstream regions, the flamelet model tends to overpredict the effect of differential diffusion which can have a profound impact on NO predictions as they are quite sensitive to temperature. As suggested by Vranos et al. [43], overprediction of differential diffusion effects is one of the possible reasons for their overprediction in NO level of a turbulent jet diffusion flame with a fuel mixture of 31% CH_4 and 69% H_2. The effect of differential diffusion on NO production now is investigated parametrically by using the assumption of unity Lewis number for all species: $\mathrm{Le}_i = \lambda/\rho c_p D_{i,\mathrm{mix}} = 1$. Here, λ is the mixture conductivity, which is taken to be $\lambda = \lambda_{N_2}$. With $\mathrm{Le}_i = 1$, all diffusion coefficients for all species are equal to a common coefficient $\mathscr{D}$. To ensure the condition $\Sigma_i Y_i V_i = 0$, the diffusion velocity V_i is evaluated using the Fickian expression $V_i = -(D/Y_i)(dY_i/dx)$. By setting $\mathrm{Le}_i = 1$, the diffusion flux of H_2 to the flame zone is reduced, which reduces the peak flame temperature and therefore the NO production rate, as presented in Fig. 3. These calculations show that the Le = 1 flamelets also exhibit a local maximum. For the full diffusion flamelets, the scalar dissipation rate at this maximum increases somewhat with increasing dilution, while for the Le = 1 flamelets, it decreases with increasing dilution. The pure hydrogen "equal-diffu-

sivity" flamelets extinguish at a strain rate of $s_q \approx 15 \times 10^3\ s^{-1}$, which is slightly higher than the value obtained for full diffusion flamelets: $s_q = 14 \times 10^3\ s^{-1}$. The corresponding scalar dissipation rates are $\chi_q \approx 130\ s^{-1}$ and $\chi_q \approx 85\ s^{-1}$, respectively. The higher extinction values of the strain rate and the scalar dissipation rate for the case Le = 1 require some explanations since the fuel flux towards the flame zone is reduced by the Le = 1 assumption. As the mixture conductivity was set equal to that of pure nitrogen, which is lower than that when H_2 contributions were included, less heat is transferred away from the flame zone, and therefore flame stability is enhanced.

The maximum value of $\dot{w}_{NO,max}$ in the equal diffusivity flamelets is lower than in the full diffusion cases for all dilutions considered. This will have a profound impact on the absolute NO level in turbulent flames. Indeed, in Fig. 2, the predictions using the equal diffusivity flamelets show lower absolute $EINO_x$ levels closer to the experimental data than the full diffusion flamelet runs. This same trend is predicted by both codes *A* and *B*. Again, the slope is close to 0.5 for large Damköhler numbers and higher for lower Damköhler numbers. However, predicted results for different dilutions appear to collapse on their own branches instead of forming one single curve as suggested by the experimental data.

Predictions with the Strain Rate

Although it has been shown analytically in Ref. 32 that the experimental 0.5 slope would not be predicted when the strain rate is used as the nonequilibrium parameter, in this section, this statement is verified by performing detailed flame calculations. The small eddy and the large eddy strain rates are defined as

$$\tilde{s}_s = 0.07\left(\frac{\epsilon}{\nu}\right)^{1/2}, \qquad \tilde{s}_l = 6\epsilon/k \tag{37}$$

where ν is the kinematic viscosity. The coefficients in these expressions are taken from Ref. 30 where they were determined from computations of turbulent lifted methane diffusion flames. Fluctuations of the strain rate are neglected, and the mean values given in Eq. 37 are associated with the strain rate at the oxidizer side of the laminar flamelets. So in each grid cell, the mean strain rate determines which single flamelet is used to compute the mean thermochemical variables. The com-

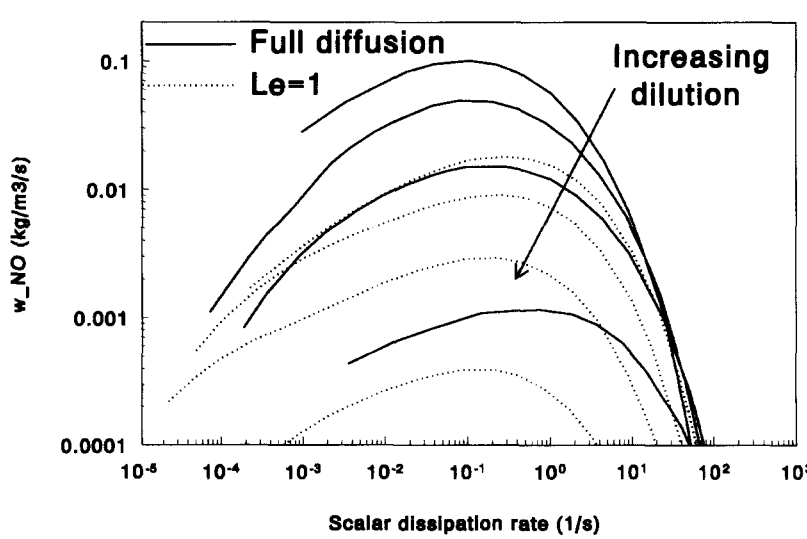

Fig. 3. Maximum NO production rate w_{NO} in kg/m^3 vs scalar dissipation rate χ_{max} in pure and diluted (0, 20, 40, and 60% He), H_2/He–air flamelets predicted with full diffusion (solid lines) and with $Le_i = 1$ (dashed lines). The curves are in descending order with increasing helium content.

puted emission indices with the small and large eddy strain rates are shown in Figs. 4 and 5, respectively. The predicted slopes with both strain rates are larger than 1, consistent with the conclusions of Ref. 32. Moreover, the predicted data do not fall on one curve; every branch corresponds to a different helium dilution. There are two major differences between the characteristics of the flamelets parameterized by the scalar dissipation rate and by the strain rate. First, the strain rate decays with downstream distance as x^{-2}, while the scalar dissipation rate decays as x^{-4}. As the flame length becomes shorter with increasing dilution, the mean scalar dissipation rate and the strain rate in the main NO production region vary differently with increasing dilution. The second difference is that the extinction values of the scalar dissipation rate increase with increasing dilution, while the values of the strain rate at extinction decrease with increasing dilution. This was also observed in Ref. 30; however, extinction has little impact on NO_x, and the second difference is probably not important to NO_x emissions.

Bray and Peters [27] argued that the scalar fluctuations the flamelet is subjected to are on the order of the flame thickness squared $(\Delta f)^2$. The effective mean scalar dissipation rate on the flamelets is then $\epsilon_f \approx 2(2f_{st})^2\epsilon/k$, using $\Delta f \approx 2f_{st}$. Predictions of code A with this model for the pure hydrogen flames are shown in Fig. 5. The results are close to the predictions using the large eddy strain rate, as in Eq. 37, showing a slope larger than 1.

A different type of flamelet model for turbulent combustion is the strained dissipation and reaction layer (SDRL) model proposed by Bish and Dahm [44]. In the SDRL model, chemical reactions are assumed to take place in locally one-dimensional strained dissipation layers. The mathematical formulation of the model is analogous to the classical flamelet model, except for the boundary values in the flamelets which do not correspond to pure fuel or pure air. This leads to a four-dimensional flamelet library with the mixture fraction, the strain rate, and the two boundary conditions as parameters. Distinct features in the SDRL model are its description of the strain rate through the scalar dissipation rate and the use of local maximum and minimum mixture fractions. To explore this alternative for prescribing the strain rate under the current modeling framework, we set the strain rate as $\tilde{s} = \epsilon_f/\tilde{f}_c^2$, and the flamelet boundary conditions are set by the

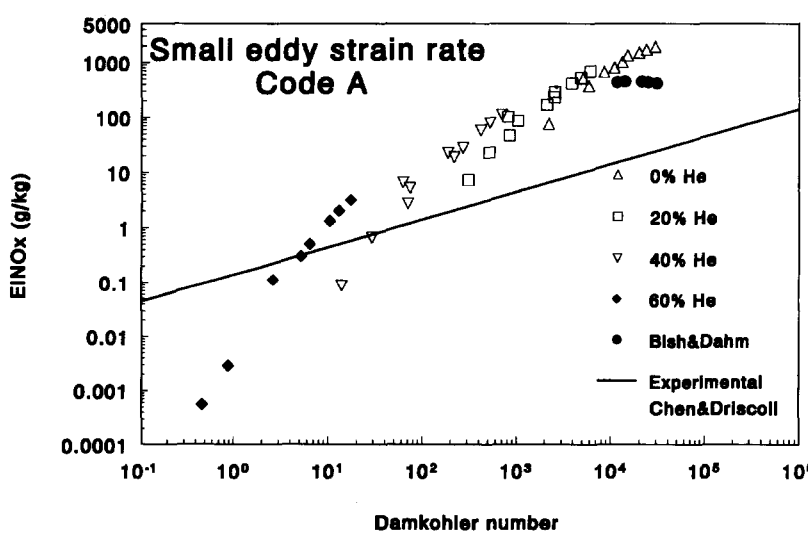

Fig. 4. Emission index vs Damköhler number, predicted with the small eddy strain rate. Predictions with the simplified model of Bish and Dahm [44] are also included. The helium dilution and the nozzle diameter (in mm) are indicated. See also legend of Fig. 2.

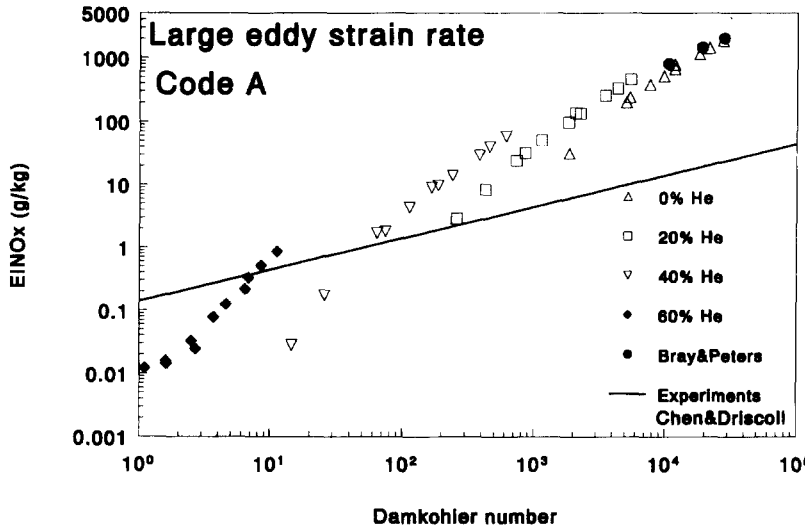

Fig. 5. Emission index vs Damköhler number, predicted with the large eddy strain rate. Predictions with the simplified model given by Bray and Peters [27] are also included. See also legend of Fig. 2.

centerline value and zero. In Fig. 4, the computed results with such a strain rate prescription show a slope of about -0.05. Since this implementation is not exactly the same as the SDRL model, future research is needed to assess the potential of SDRL in predicting NO_x emissions from turbulent jet diffusion flames.

CONCLUSIONS

The present study on flamelet modeling reveals that when the nonequilibrium parameter is assigned by the scalar dissipation rate, the predicted $EINO_x$ vs Da curves collapse onto one curve, as observed in the experiments of Chen and Driscoll [1]. Although the model overpredicts the NO_x emission levels, the predicted slope of $EINO_x$ vs Da on a logarithmic plot is close to the experimental value of 0.5 at large Da numbers. The flamelet predictions start to deviate from the 0.5 slope for small Damköhler numbers, consistent with the notion that the flamelet approach is appropriate for high Damköhler number turbulent flames. The overpredicted NO_x levels are likely attributed to the overpredicted differential diffusion effect by the flamelet model. This has been shown by artificially setting all diffusitivities equal to a common value based on unity Lewis number. This lowered the absolute $EINO_x$ significantly, while preserving an approximately correct slope for large Damköhler numbers. Different dilutions, however, seem to generate different branches, which is not in agreement with the experiments. More research is needed to understand the implications and utility of setting Le = 1 in the flamelets.

Using the small-scale or the large-scale strain rate as the nonequilibrium parameter gives much too high values for the slope, and moreover, the predicted results do not collapse onto one curve. When compared to other turbulent combustion methods that reproduce the experimental 0.5 slope, the current flamelet model gives the largest deviation regarding absolute $EINO_x$ values. This has been shown to be caused by the differential diffusion effects, which will be investigated in more detail in future work.

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